AI ACADEMY · LEVEL 1 · GRADE 6 · WEEK 15 · STUDENT LESSON

Save New Examples for Testing

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Helpful Classroom Sorter

THE CONNECTED WEEK

Freeze the model. Keep the first result.

Separate development and evaluation, inspect completed experiments and run a new versioned envelope protocol.

Keep one habit: Reserve first → freeze and predict → reveal and report without rewriting history.

MAIN LESSON · SECTION 1

When is a test still independent?

> Record a frozen model and predictions before labels are revealed, then describe any revision honestly.

Start with an improved replay

A model can improve after feedback while its new score becomes a development result. Keep the original frozen model's predictions so the first test is not rewritten after the revision.

Retrieve validation versus correctness

A height passes the **input** contract and the model predicts tall. What is still needed to check that **prediction**? If its comparison label changes your threshold choice, what role has that example played?

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

By the end, you can...

- I can separate examples used to choose a model from examples reserved for its final evaluation.
- I can record a fixed model and its **prediction**s before revealing comparison labels.
- I can explain how test feedback changes the **evidence** available for a revised model.

▼ **Earlier wording, alternatives and picture transcripts**

Your learning target

Remember before reading

Without opening the old lesson, explain one key idea from Build a Human Classifier. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Debug the Cause. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Algorithms as Recipes. Give an example and a mistake that the idea helps you avoid.

A fair test uses examples not used to create the decision **process**.



LEARNING TARGET Eight cards are used to invent a rule. Four sealed cards are then opened; the team changes its rule after seeing errors. Can those four cards still be called an untouched final test? Plan what to do next.

Learner name:

Readiness for the connected explanation

This week’s end goal

By the end, I can **freeze** a model before labels are revealed and report a bounded test result.

MAIN LESSON · SECTION 2

Separate development from evaluation before looking

> The protocol fixes the model before reserved inputs and their labels are used to judge it.

Reserve examples before development

A rule chosen to fit examples may work differently on new cases. To investigate that difference, reserve a set of examples before development. Use the **development examples** to choose the model.

Keep the final test examples separate until the model is fixed, then evaluate that recorded model.

Freeze the model and preserve predictions

In this paper activity, a teacher keeps the reserved examples in an envelope. Record the **input** contract, feature, category rule and model version before opening it.

Once the **input** values are revealed, predict using the frozen model before revealing their comparison labels. This prevents copying the labels into the **prediction**s.

Distinguish a completed demonstration from your new experiment

The worked example below describes a completed experiment, so its test labels are visible for discussion. It is not an unseen test for you.

In the workbook experiment, the teacher will supply a separate set of comparison labels after you have saved your **prediction**s. All data are imaginary classroom records.

Explain the order of evidence

- What must be recorded before the envelope opens? What must be saved before the comparison labels are revealed? Explain why the visible worked example is practice for you rather than an unseen test.

▼ Earlier wording, alternatives and picture transcripts

PICTURE STORY

Look Closely Before You Explain

Pictures give clues; **evidence** tells us what the clues mean.



MYSTERY If a sorter sees every mystery card while its rule is being created, later success on those same cards does not show transfer.

What do you notice?

Name three visible details without explaining them yet.

Circle the detail most connected to the system's decision.

Name one detail that may be decoration or distraction.

Make a first **prediction** and state the clue that supports it.

My **observation**s, **prediction**, and **evidence**:

GOOD THINKING A **prediction** can change when stronger **evidence** appears.

A closer explanation

Freeze, predict, reveal and compare

> The model version links each result to the procedure actually evaluated.

Read the three evidence stages

Different roles for development and final testing

EVIDENCE AVAILABLE	ALLOWED STEP	RECORD TO KEEP
Development examples	Fit and select a model	Choice and its reasons
Reserved inputs; labels hidden	Use the frozen model	Predictions and version
Reserved labels revealed	Compare without revising	Results for that version

Read the rows in order. Information from the final comparison must not travel back into model selection while the result is still described as an independent final test of that model.

Identify the completed experiment’s frozen TowerA

In the completed experiment, development towers at heights 2, 4, 6 and 8 cm have labels short, short, tall and tall. Candidates predict tall at or above 5 or 6 cm.

Both fit all four labels; the stated smaller-threshold tie rule selects 5. The team **freeze**s model TowerA with threshold 5.

Save predictions on the reserved inputs

The reserved **input**s are different records at heights 3, 5, 7 and 9 cm. TowerA accepts whole cm heights from 1 through 10. It predicts short, tall, tall and tall.

These **prediction**s are saved before the reserved labels are opened. The comparison below records what happened, including the failure.

Explain a stage transition

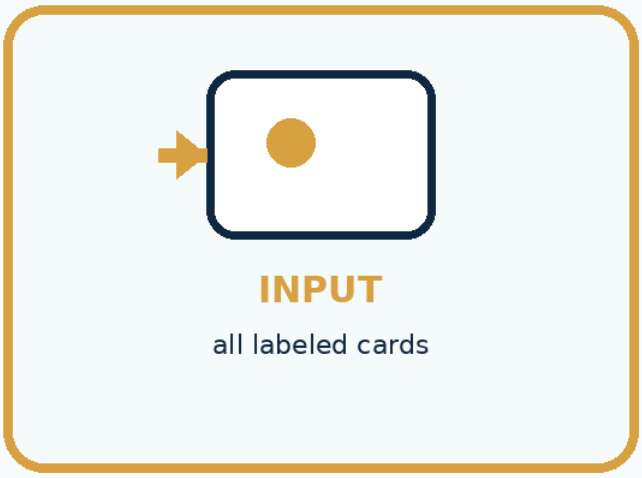
- Point to the **input**, the deciding step and the result in this week's visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

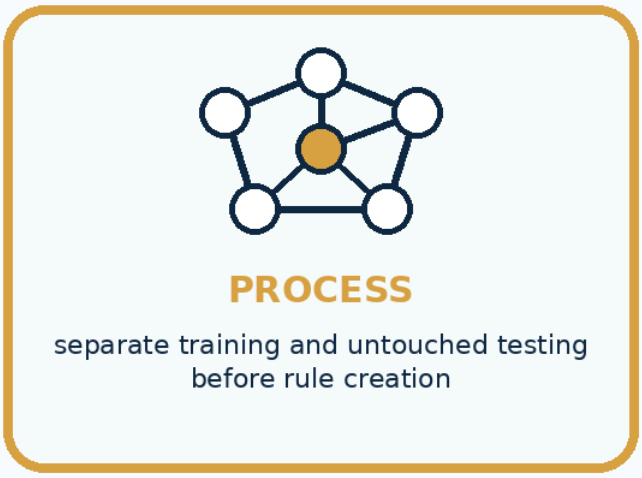
▼ Earlier wording, alternatives and picture transcripts

VISUAL **MECHANISM**

Input → Process → Output

Follow the information instead of imagining magic inside the machine.







INPUT all labeled cards

PROCESS separate **training** and untouched testing before rule creation

OUTPUT honest performance **evidence** on new cards

CORE IDEA A fair test uses examples not used to create the decision **process**.

UNDER THE HOOD Evaluation on **training** data is optimistically biased. A **holdout test set** estimates behavior on unseen examples only if it remains untouched during design.

Repeatedly tuning to test results leaks information and turns the test into **training**.

Draw a fresh example with all three stages:

A closer explanation

Connect the picture and the explanation

MAIN LESSON · SECTION 4

Compare development, evaluation and feedback use

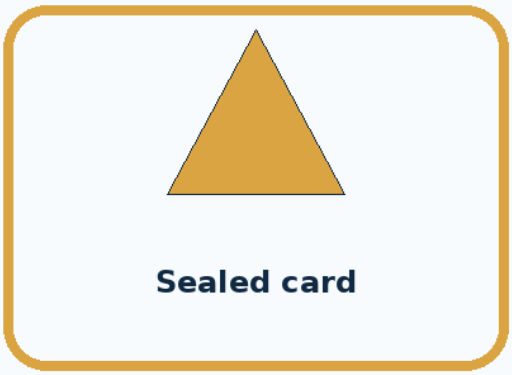
> Ask which examples or scores influenced a model choice before calling a result an independent test.

Pictures paired with their evidence



The case to investigate

CASE 1 — Card used to write rule Clue: belongs in learn pile. Meaning: **training evidence**.



The case to investigate

CASE 2 — Sealed card Clue: unseen until final run. Meaning: test **evidence**.



The case to investigate

CASE 3 — Peek then revise Clue: test influenced **process**. Meaning: leakage.



The case to investigate

CASE 4 — Fresh local card Clue: new after design. Meaning: strong transfer check.

Discuss the confusing pair

- Which two pictures are easiest to confuse? What **evidence** separates them?

Keep the limit in view

BOUNDARY A **test set** stops being independent when its answers guide revisions.

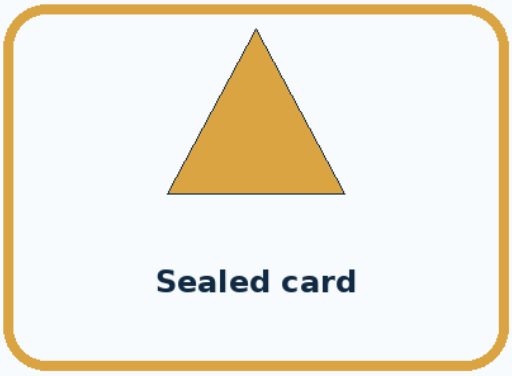
▼ Earlier wording, alternatives and picture transcripts

PICTURE GALLERY

Four Cases, Four Clues

Use the visual cue, then read the **evidence** carefully.









CASE 1 — Card used to write rule Clue: belongs in learn pile. Meaning: **training evidence**.

CASE 2 — Sealed card Clue: unseen until final run. Meaning: test **evidence**.

CASE 3 — Peek then revise Clue: test influenced **process**. Meaning: leakage.

CASE 4 — Fresh local card Clue: new after design. Meaning: strong transfer check.

Change one thing in the pictured case

Change: After feedback, change the boundary to 5.

Prediction: The revised rule matches all four known labels but needs a different final test.

Reason: These labels influenced the revision, so replaying them is **evidence** of repair rather than unseen performance.

MAIN LESSON · SECTION 6

Keep the original three-of-four result

> This completed PencilP demonstration has visible labels for discussion; it is supported practice.

The frozen model, reserved inputs and completed label record

A fictional envelope exercise evaluates a fixed pencil-length sorter.

Freeze PencilP: long at least 6 cm; short below 6.

Reserved valid lengths are 3,5,6,9 cm. Teacher envelope labels, revealed only after **prediction**s, are short,long,long,long.

Compare saved predictions with revealed labels

AI ACADEMY / AI-1/15

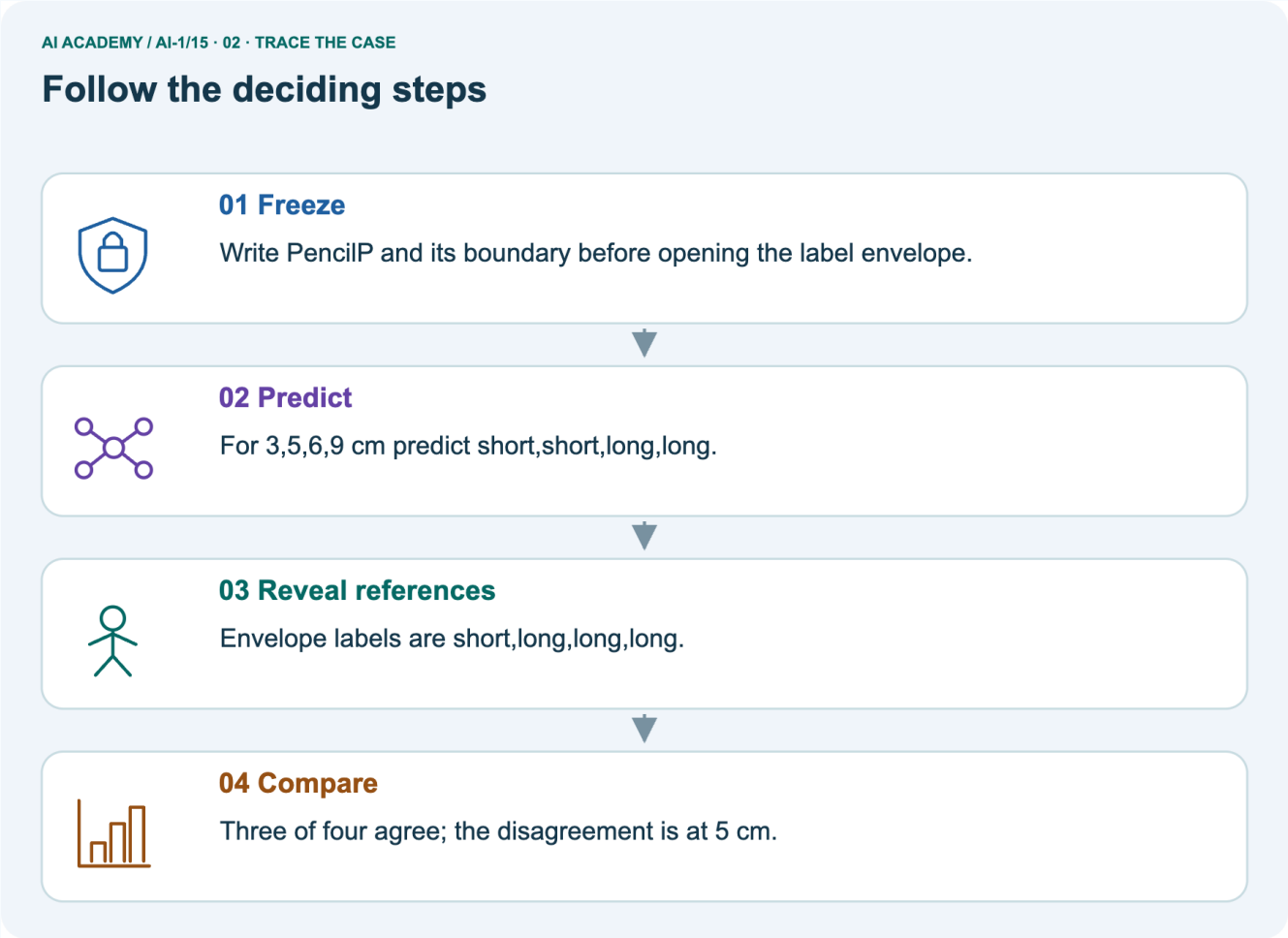
Frozen predictions versus reserved labels

	Length cm	PencilP	Reference	Result
Test 1	3	short	short	agree
Test 2	5	short	long	disagree
Test 3	6	long	long	agree
Test 4	9	long	long	agree

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

Trace the evaluation sequence



Follow the labelled picture with the matching explanation. Open the vector diagram

State the bounded 3/4 result

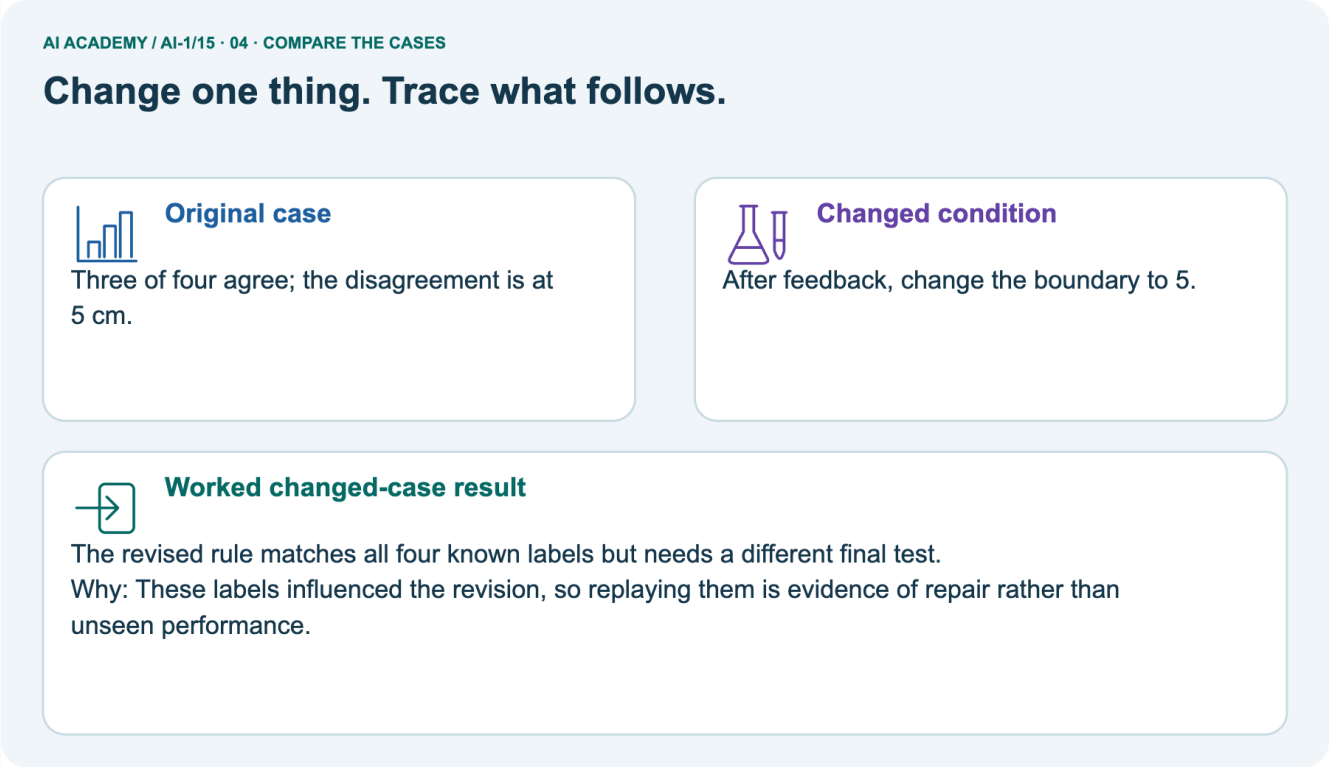
The **prediction** record must exist before the labels are seen. PencilP scores 3/4=75% agreement on these four selected cases.

Moving the boundary after inspecting the envelope makes this a development exercise; the same four cases can no longer serve as untouched final evaluation.

Predict after the feedback-guided revision

- Before checking, revise PencilP's boundary from 6 to 5 after the visible feedback. Predict the replay labels. Explain why its result on these already-seen records has a different evidence role from the original frozen prediction record.

Check the replay and its evidence role



Follow the labelled picture with the matching explanation. Open the vector diagram

Limit the claim beyond four selected cases

Four selected cases cannot establish a reliable accuracy rate for every future pencil or classroom.

Explain why a replay does not replace the original

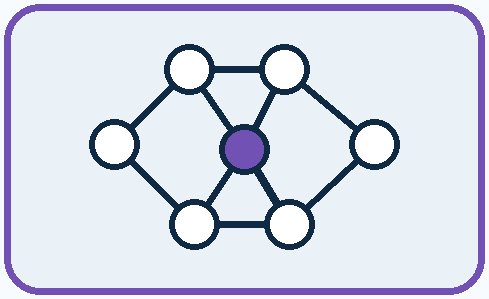
- Read the labelled picture: Does 4/4 on the replay replace the original 3/4 result? Explain.

▼ Earlier wording, alternatives and picture transcripts

WORKED EXAMPLE

Watch the Reasoning, Not Just the Answer

Every step connects a claim to observable **evidence**.



CASE If a sorter sees every mystery card while its rule is being created, later success on those same cards does not show transfer.

STEP 1 — OBSERVE all labeled cards

STEP 2 — TRACE separate **training** and untouched testing before rule creation

STEP 3 — CONCLUDE honest performance **evidence** on new cards

MODEL REASONING Randomly or deliberately reserve representative cards first. Create the decision **process** using only the learning pile.

Open the test pile once for **evidence**, record every result, and do not call the revised **process** tested until a new **holdout** exists.

DO NOT OVERCLAIM A **test set** stops being independent when its answers guide revisions.

What new **evidence** could change this answer?

A closer explanation

Reserved height	Saved TowerA prediction	Revealed comparison label
3 cm	Short	Short
5 cm	Tall	Short
7 cm	Tall	Tall
9 cm	Tall	Tall

TowerA agrees on three of these four reserved examples and disagrees at 5 cm. That is **evidence** about this fixed model on this small set. It does not prove a success rate for every future tower.

The labels, range of examples and way records were selected also matter.

After seeing the 5 cm label, the team changes the threshold to 6 and names the revision TowerB. TowerB agrees on all four of these now-seen examples.

The change may be useful, but those examples have influenced its development. Four agreements on them are not an independent final test result for TowerB. Keep TowerA's original three-of-four record.

Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

A pictorial worked case: Save New Examples for Testing

The facts we will use

AI ACADEMY / AI-1/15

Frozen predictions versus reserved labels

	Length cm	PencilP	Reference	Result
Test 1	3	short	short	agree
Test 2	5	short	long	disagree
Test 3	6	long	long	agree
Test 4	9	long	long	agree

Read a cell using both its row and column labels.

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-1/15

Save New Examples for Testing

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Freeze

Write PencilP and its boundary before opening the label envelope.



2. Predict

For 3,5,6,9 cm predict short,short,long,long.



3. Reveal references

Envelope labels are short,long,long,long.



4. Compare

Three of four agree; the disagreement is at 5 cm.

CHANGE ONE THING

After feedback, change the boundary to 5.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Freeze	Write PencilP and its boundary before opening the label envelope.
2. Predict	For 3,5,6,9 cm predict short,short,long,long.
3. Reveal references	Envelope labels are short,long,long,long.
4. Compare	Three of four agree; the disagreement is at 5 cm.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

Why this result follows

What this example does not establish

Change one thing: After feedback, change the boundary to 5. Predict the result before checking the explanation. What remains the same?

MAIN LESSON · SECTION 7

Unseen is necessary but coverage also matters

> Copies and easy near-duplicates can make an apparently reserved set weak evidence.

Inspect how records were selected

Keeping a set unseen is necessary for this kind of independent check, but secrecy alone does not make it a good test. Four near-identical easy cases provide little **evidence** about varied cases.

Copies of development records can also make a reserved set misleading even if their file names differ.

Plan for the intended use and retained outcomes

Define what future cases the model is meant to handle and choose evaluation examples that address that use. Keep related or copied records together when separating sets.

Record invalid **input**s and no **prediction** outcomes separately instead of quietly dropping them from the report. Later weeks examine coverage and performance comparisons in more detail.

Repair the best-of-ten-test-scores claim

- CLAIM A Testing on familiar cards is fair because there are more correct answers.
- CLAIM B Looking at test cards without labels cannot affect design.

STRONG ANSWER FRAME I claim _____ because the **evidence** shows _____. The **process** is _____. One limit is _____.

> **A team tries ten thresholds, reads the reserved test score for each, and reports only the best score as its final unseen test. Explain the problem and give an honest description of what it has measured**

▼ **Earlier wording, alternatives and picture transcripts**

COMPARE

Same Result, Different Reason

A comparison helps reveal the true concept boundary.

Case	Evidence	What it shows
Card used to write rule	belongs in learn pile	training evidence
Sealed card	unseen until final run	test evidence
Peek then revise	test influenced process	leakage
Fresh local card	new after design	strong transfer check

Repair two tempting claims

Repair:

Repair:

A closer explanation

MAIN LESSON · SECTION 8

Practise: report both model versions honestly

> A revised model may use feedback, but it needs fresh appropriate evidence for a new independent evaluation.

Keep improvement and evaluation roles separate

Improvement after a test is allowed. State that the old test informed the revision, preserve both versions, and reserve fresh appropriate examples to evaluate the revised model independently.

If none are available, report that limitation instead of inventing a fresh test result.

> **Write two result statements: one for TowerA on the original reserved set, and one for TowerB after the revision. Explain what must happen before a new independent evaluation of TowerB. Would shuffling the four old records make them unseen again**

Recognise selection through repeated test scores

A test can lose independence even when you never copy its labels into a **training** table. Repeatedly trying thresholds and choosing the one with the best test score also uses test information to choose the model.

Separate development comparisons from the final check.

Use W2 to preserve the original result

Use workbook W2 for the connected example “Keeping final tests unseen”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

Plan	Do	Check
What is given? Which rule or procedure applies?	Record each intermediate step.	Does the result follow? What remains unknown?

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

Challenge “a new file makes records unseen”

- Use the worked case for Keeping final tests unseen. Proposed explanation: “A new file makes old records unseen”. Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

- Answer on your own before discussion. If you are unsure, say which step you need help with.

Explain the pictured replay

Does 4/4 on the replay replace the original 3/4 result? Explain.

My explanation using a pictured step and the given facts

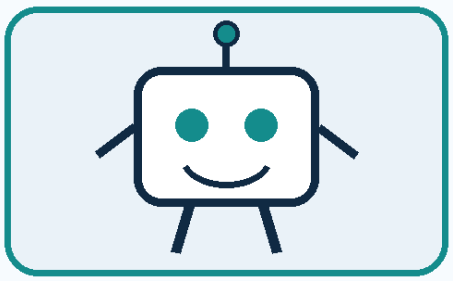
This is supported practice. Keep the separately named independent case for an attempt before feedback.

▼ **Earlier wording, alternatives and picture transcripts**

GUIDED PRACTICE

Try It, Check It, Explain It

Use support now so you can succeed independently later.



Explain how the picture story demonstrates this idea: A fair test uses examples not used to create the decision **process**.

Use a fresh example to explain **training set**.

Trace a new **input** → **process** → **output** example.

Find and repair the error in: Looking at test cards without labels cannot affect design.

Invent one edge case and name the safe response.

PARTNER CHECK Point to the **evidence**. Do not accept an answer that only repeats the vocabulary word.

A closer explanation

Practise with less help: Preserve the original result

Check your reasoning

Explain the picture

MAIN LESSON · SECTION 9

Investigate a learn/test envelope protocol

> Use supplied development records, a declared selection policy and teacher-held comparison labels.

Development cards and model-selection rules

Use imaginary rod records. Development examples are 1 cm=SHORT and 5 cm=LONG. Candidate boundaries 3 and 4 predict LONG at or above the boundary and SHORT otherwise. Choose fewer disagreements; a tie prefers the larger boundary. Accept whole centimetres 1–6. Before opening the reserved-input envelope, record the selected rule and version RodD.

Freeze, predict, reveal and preserve

1. The separate reserved inputs are 2,3,4,6 cm. Save all four RodD predictions before your teacher reveals their comparison labels from the guide. Compare once and keep the original record. If you then revise the boundary using that feedback, name a new version and describe its replay as development evidence. Keep this supported exercise separate from the independent RibbonA experiment.

2. Keep a normal case and an edge case clearly separated.

3. Record what happened—even when it does not match your **prediction**.

4. Explain the result, one limitation, and the next useful test.

Keep versions and evidence roles

Case	Prediction	Observed result	What it means
Normal			
Edge			

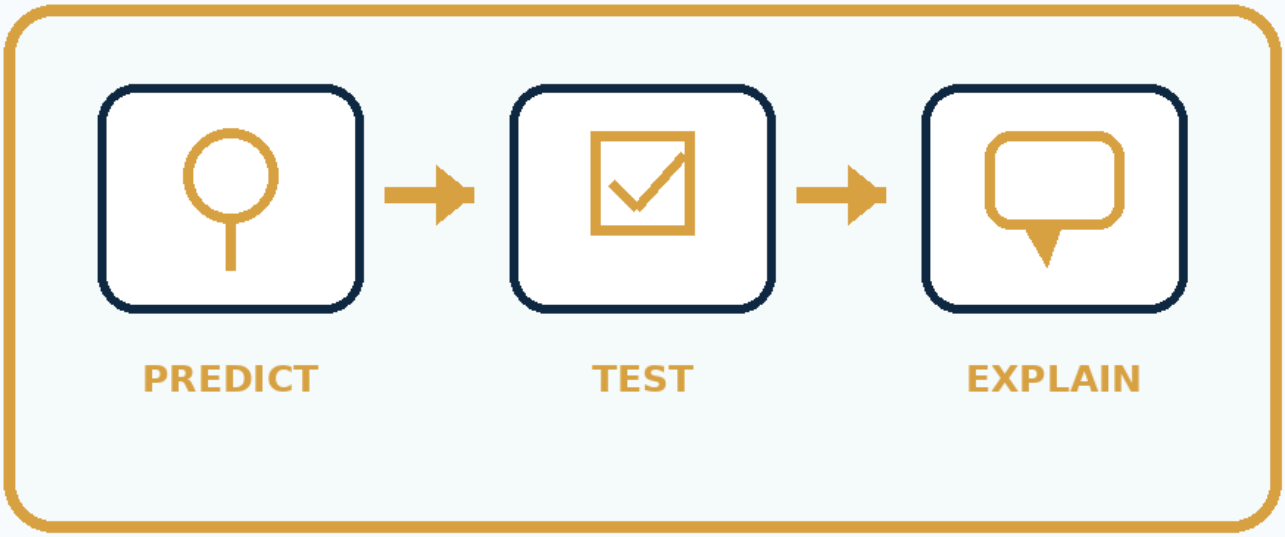
Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

▼ Earlier wording, alternatives and picture transcripts

EXPLORER LAB

Learn/Test Envelope Split

Predict first, test honestly, and explain what the **evidence** means.



State your **prediction** before the result is known.

Complete the task: Split card data into learn and test piles before sorting.

INDEPENDENCE After support is removed, repeat the proof with a fresh case.

Plan → try → check

MAIN LESSON · SECTION 14

Try alone: run the separate RibbonA experiment

> Freeze RibbonA and save its predictions before asking for the teacher-held comparison labels.

The new protocol and reserved inputs

Before receiving any test labels, **freeze** RibbonA: whole cm lengths 1 through 12, long at least 6 cm and short otherwise. Save **prediction**s for 5, 7, 9 and 11 cm.

Ask your teacher to reveal the reserved labels, then record agreements, disagreements and a limited result statement.

If you later choose threshold 8 after seeing these labels, explain how you must describe its result on the same records.

> **My independent explanation, trace and limits**

Keep one first attempt

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Your teacher has the independent answer and scoring criteria. Complete the matching workbook week to keep your practice **evidence** together.

Use feedback afterwards

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

▼ Earlier wording, alternatives and picture transcripts

Independent case: Run a separate ribbon experiment

Week 15 Independent application

Close the worked explanation. Apply the idea to this different question without copying.

You teach a sorter using cards A, B and C. You keep D and E unseen. Which cards provide the fairer first test? What changes if you show D during teaching?

My answer, with a trace, calculation or supporting **evidence**

Write or draw your reasoning and evidence.

Create and test

Change one relevant detail. Predict the result before testing. Include a boundary or missing-information case when it helps reveal a limit.

My changed **input** and expected result

Write or draw your reasoning and evidence.

Connect to your project

Create separate learn and test envelopes for the Classroom Sorter and label who may open each.

The **evidence** I will keep

Write or draw your reasoning and evidence.

Transfer practice from the original programme

Eight cards are used to invent a rule. Four sealed cards are then opened; the team changes its rule after seeing errors. Can those four cards still be called an untouched final test? Plan what to do next.

Explain the mechanism, evidence and one limit.

Use feedback to improve

MAIN LESSON · SECTION 11

Reserve project cases before using their answers

> Keep dated sorter versions and a record of which examples shaped each revision.

This week’s project decision

THIS WEEK'S CHECKPOINT Create separate learn and test envelopes for the Classroom Sorter and label who may open each.

Maintain dated sorter versions and separate **development examples** from reserved project-evaluation cases.

Plan envelopes, predictions and review

- What real classroom need does this serve?
- What information is necessary—and what should stay private?
- What happens inside the **process**?
- What normal and edge cases will you test?

- When should it say not sure or ask a person?

Keep test feedback's influence visible

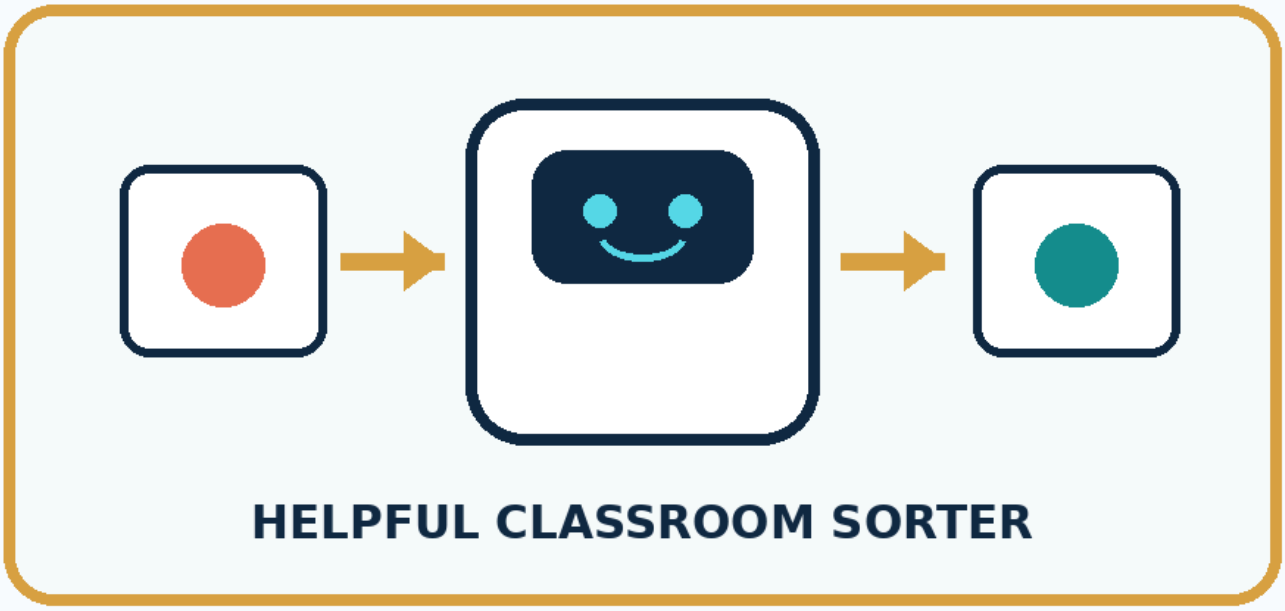
RESPONSIBLE DESIGN A **test set** stops being independent when its answers guide revisions.

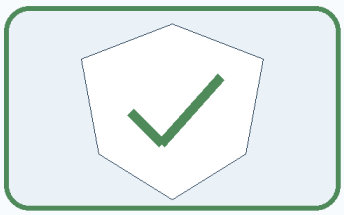
▼ Earlier wording, alternatives and picture transcripts

PROJECT

Helpful Classroom Sorter

Build one coherent system across the year, one **evidence**-based decision at a time.





My project decision:

Evidence for your project

MAIN LESSON · SECTION 16

Finish: name the model and the role of its score

> Close your notes and explain what was reserved, what was revised and what remains untested.

Your exit explanation

- Close your notes. Explain this goal in your own words: Separate examples used to choose a model from examples reserved for its final evaluation

> **My deciding step, evidence and one remaining question**

Choose support and revisit later

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Your quick word guide**

REFERENCE · SECTION 5

Your quick word guide

> Use the words to explain a particular step, record or limit.

Core vocabulary

Word	Meaning	My example
training set	examples used to create or adjust a process	
test set	untouched examples used for final checking	
holdout	data deliberately kept separate	
data leakage	test information influences design	

Terms for the connected explanation

Word	Meaning
Development examples	Examples available for fitting, comparing or revising a model
Held-out test set	Examples reserved from development for evaluating a fixed model
Freeze	Record a model and stop changing it for a particular evaluation

Word	Meaning
Test leakage	Information from a reserved test influencing model development
Evaluation record	A record of the tested model, predictions, comparison labels and results

Repair a claim using a word

- CLAIM TO REPAIR Testing on familiar cards is fair because there are more correct answers.
- Use one vocabulary word and one piece of **evidence** to repair it:

▼ Earlier wording, alternatives and picture transcripts

VOCABULARY

Words You Can See and Use

The badge is a memory cue; the definition does the real thinking work.









Misconception detective

WORD POWER A strong definition helps you decide something in a new case.

Words used in the closer explanation

▼ Choose a stronger thinking challenge

REFERENCE · SECTION 10

Choose a stronger thinking challenge

> Choose the next useful challenge rather than treating the whole ladder as compulsory.

Pick your next step

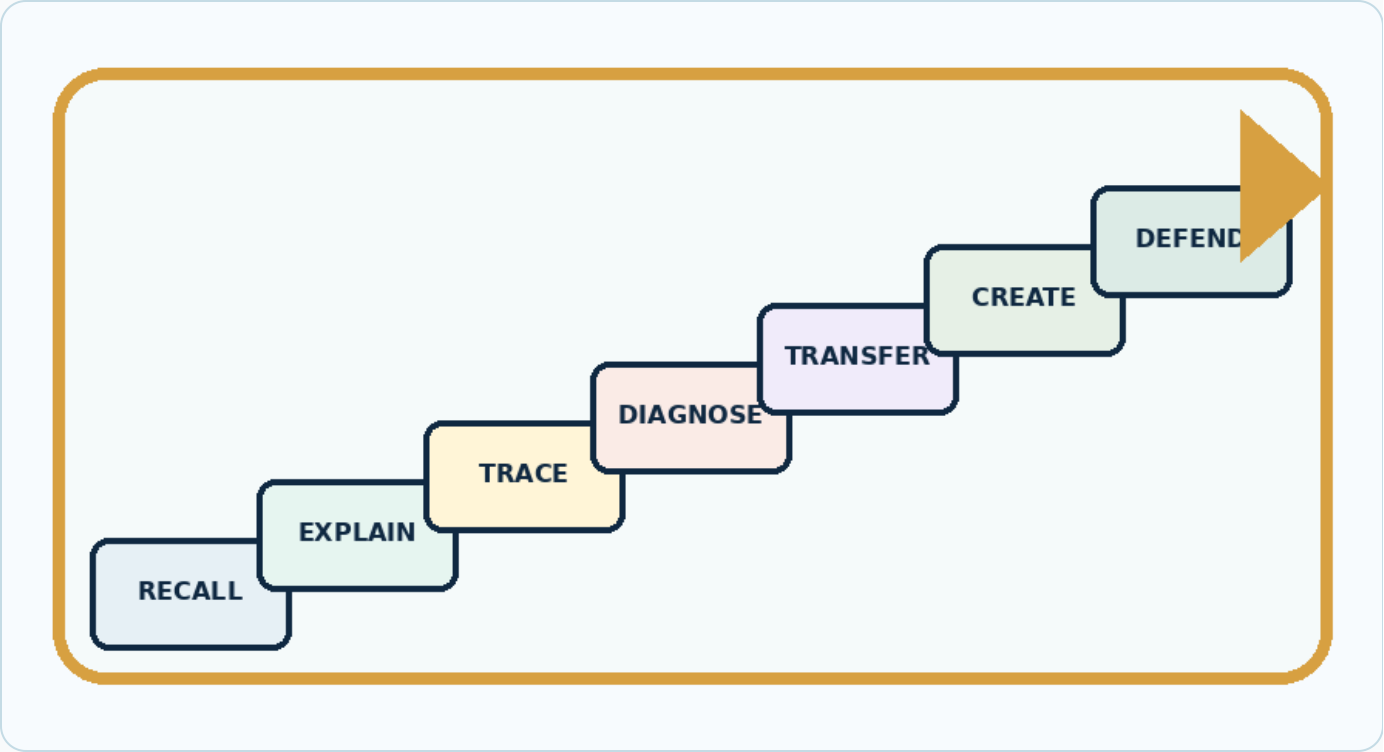
- RECALL Define **training set**.
- EXPLAIN A fair test uses examples not used to create the decision **process**.
- TRACE Show the **input**, **process**, and **output**.
- DIAGNOSE Repair: Testing on familiar cards is fair because there are more correct answers.
- TRANSFER Apply the idea to a new home or classroom case.
- CREATE Design one revealing test.
- DEFEND Give a claim, **evidence**, **mechanism**, limit, and safe action.

▼ Earlier wording, alternatives and picture transcripts

PRACTICE LADDER

Climb from Recall to Defense

Each step asks for a stronger kind of understanding.



RETRY RULE A coached answer becomes mastery only after you solve a different case independently.

▼ **Reference: check your explanation**

REFERENCE · SECTION 12

Reference: check your explanation

> Use after practice. Run a separate ribbon experiment is the selected independent case.

Check the evidence and its limit

- ☐ Name or draw the fresh case.
- ☐ Trace **input**, **process**, and **output**.

- ☐ State the conclusion and strongest **evidence**.
- ☐ Name one limit, failure, missing clue, or fairness/privacy concern.
- ☐ Explain one safe next action or improvement.

Record support

My level: ☐ Not yet ☐ With one hint ☐ Independently ☐ I can teach it

▼ **Earlier wording, alternatives and picture transcripts**

MASTERY

Checklist for explaining this week's idea

Use after practice. Run a separate ribbon experiment is the selected independent case.



EXPLANATION CHECK Freeze the version, save predictions before labels, retain the original result and explain when test feedback becomes development evidence.

Independent **evidence**:

SELF-CHECK Randomly or deliberately reserve representative cards first. Create the decision **process** using only the learning pile.

Open the test pile once for **evidence**, record every result, and do not call the revised **process** tested until a new **holdout** exists.

NEXT BRIDGE Carry today's **evidence** habit into: Boundaries and 'Not Sure'.

▼ **Extra practice: Keep the final cards unseen**

REFERENCE · SECTION 13

Extra practice: Keep the final cards unseen

> Try the question before opening its worked explanation.

Read the supplied case and question

Cards A–F are used for practice. G and H remain sealed.

Which can provide a fresh test, and under what condition?

Your explanation

- Which detail supports the answer, and why does it matter?

Write or draw your reasoning and evidence.

▼ Check after trying

G and H, only if neither influenced teaching or revision. Familiar cases do not establish unseen performance. Keep final test answers separate.

Evidence or stage	Value or result
Practice A–F	6
Sealed test G–H	2

▼ Earlier wording, alternatives and picture transcripts

Week 15 Worked case

Use the supplied details to practise the weekly concept before attempting the independent question. This case extends the illustrated lesson.

The situation

The question

Worked explanation

Explain it in your own words

Trace the supplied case visually

Keep the final cards unseen

Cards A–F are used for practice. G and H remain sealed.

Which can provide a fresh test, and under what condition?

G and H, only if neither influenced teaching or revision. Familiar cases do not establish unseen performance. Keep final test answers separate.

▼ Reference: original lesson checklist

REFERENCE · SECTION 15

Reference: original lesson checklist

> Use an additional checklist when it helps inspect your explanation.

Targets and evidence

By the end, I can	How I will show it
Separate examples used to choose a model from examples reserved for its final evaluation	Explain the deciding step, show a trace or test, and state one limit.
Record a fixed model and its predictions before revealing comparison labels	Explain the deciding step, show a trace or test, and state one limit.
Explain how test feedback changes the evidence available for a revised model	Explain the deciding step, show a trace or test, and state one limit.

Keep your route and first prediction

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

▼ Earlier wording, alternatives and picture transcripts

Start with this question: A fictional envelope exercise evaluates a fixed pencil-length sorter.

▼ Extra practice: pictures, repair and feedback

REFERENCE · SECTION 17

Extra practice: pictures, repair and feedback

> Attempt each question before opening its explanation. Use one browser or paper copy.

First supported question

- A fictional envelope exercise evaluates a fixed pencil-length sorter.
- Which labelled step matches this detail? Three of four agree; the disagreement is at 5 cm.
- 1. Reveal references
- 2. Compare
- 3. **Freeze**
- 4. Predict
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ Check the first explanation

After your attempt, compare with the supported explanation: Compare Three of four agree; the disagreement is at 5 cm. The **prediction** record must exist before the labels are seen.

PencilP scores 3/4=75% agreement on these four selected cases. Moving the boundary after inspecting the envelope makes this a development exercise; the same four cases can no longer serve as untouched final evaluation.

Changed-case question

- Changed-case practice: After feedback, change the boundary to 5. Which conclusion is supported by the supplied facts?
- 1. The revised rule matches all four known labels but needs a different final test.
- 2. We can decide the result without examining the changed **input** or the procedure.
- 3. This one example guarantees correct results for every future **input**.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

> **My selected conclusion, deciding fact and explanation**

▼ Check the changed case

After your attempt, compare with the supported explanation: The revised rule matches all four known labels but needs a different final test.

These labels influenced the revision, so replaying them is **evidence** of repair rather than unseen performance.

Repair and save your work

Review Build a Human Classifier (ai-1/14). Goal: Construct a classifier with a stated **input** format, selected rule and recorded **output**.

Short practice: Trace 0 cm and explain why clarification is different from predicting short.. Return to this week and explain what you changed.

Review Debug the Cause (ai-1/12). Goal: Compare expected and observed traces to locate the first difference. Short practice: Propose **input** 0 and give both predicted **output**s without inventing an observed result..

Return to this week and explain what you changed.

Review Algorithms as Recipes (ai-1/9). Goal: Trace an ordered algorithm using the current value after every instruction. Short practice: Use the arrows to explain where the two zero-start runs first differ..

Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

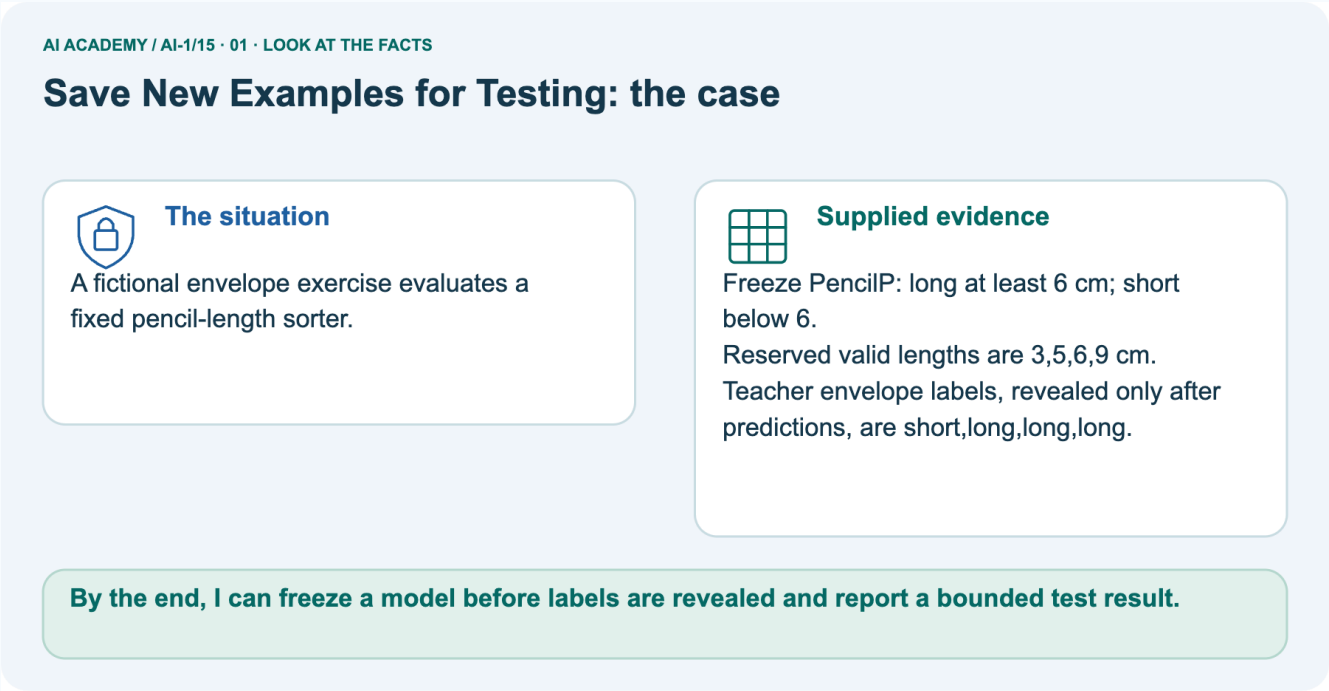
Keep your completed work

▼ Reference: the six-picture journey

Reference: the six-picture journey

> Revisit the picture that serves the current question; key diagrams already sit beside the main explanation.

01. Save New Examples for Testing: the case



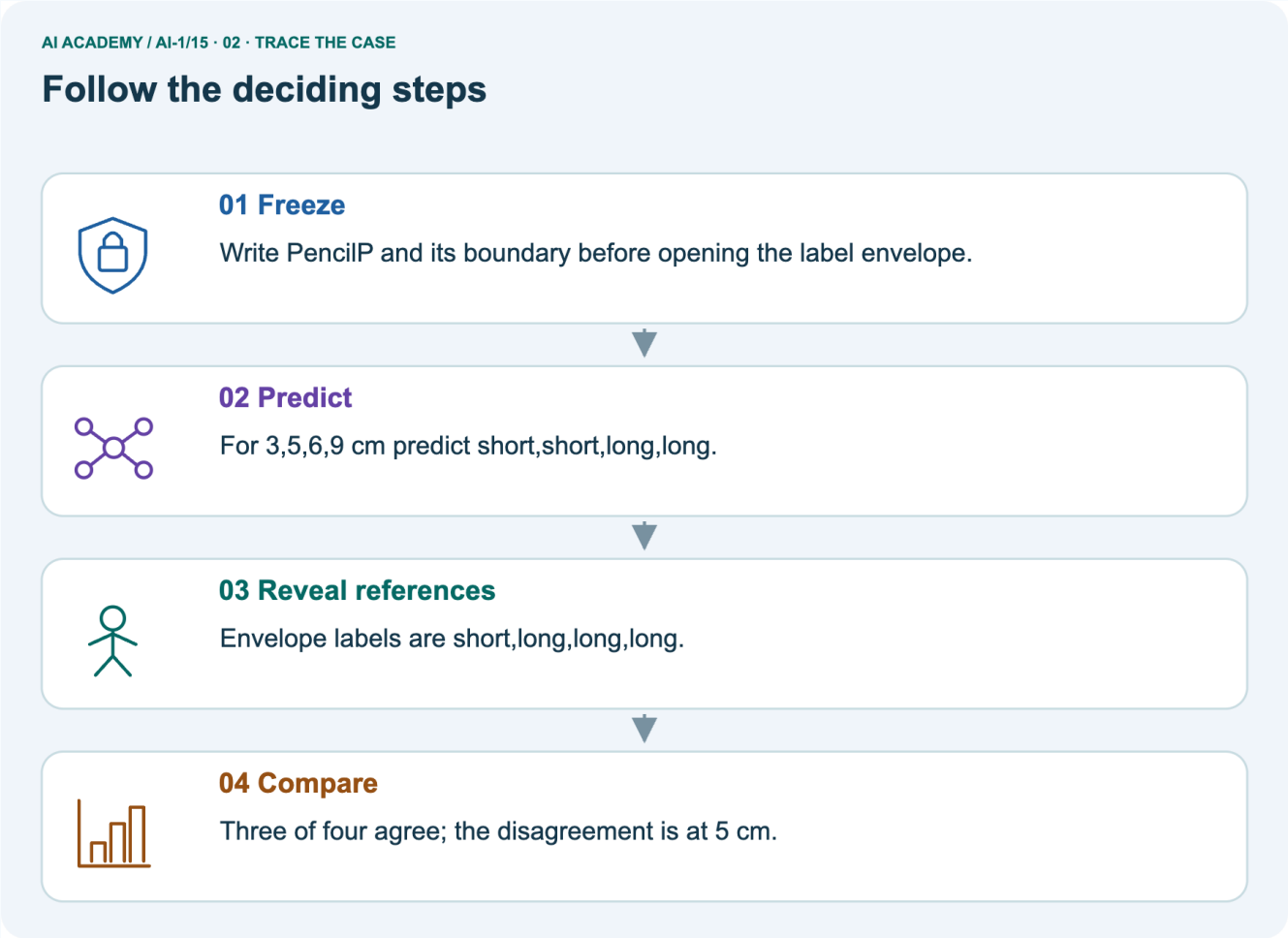
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional envelope exercise evaluates a fixed pencil-length sorter. Supplied facts: **Freeze** PencilP: long at least 6 cm; short below 6. Reserved valid lengths are 3,5,6,9 cm.

Teacher envelope labels, revealed only after **prediction**s, are short,long,long,long.

Point to the given fact that matters most. Explain why it matters.

02. Follow the deciding steps

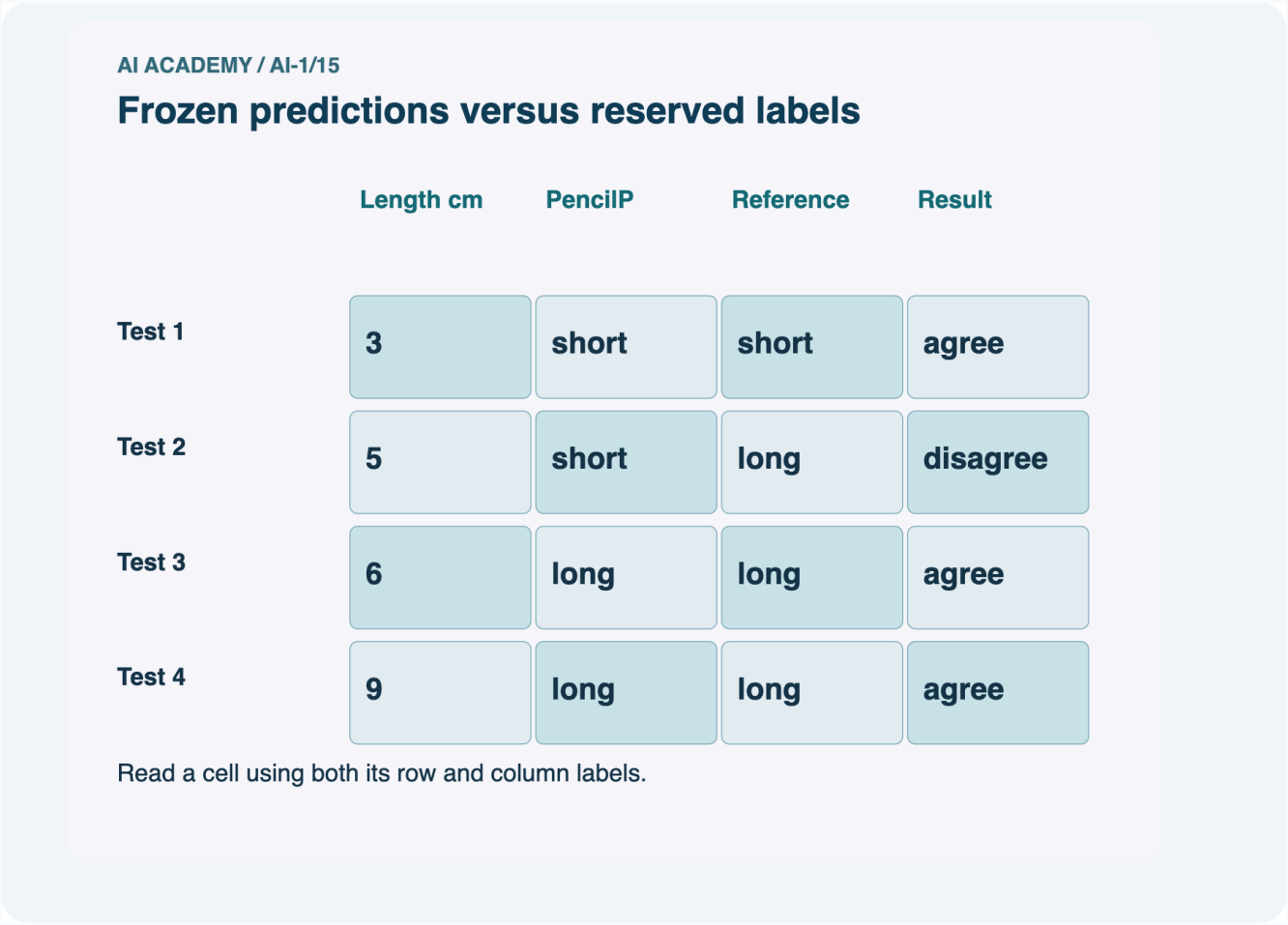


Follow the labelled picture with the matching explanation. Open the vector diagram

Freeze: Write PencilP and its boundary before opening the label envelope. → Predict: For 3,5,6,9 cm predict short,short,long,long. → Reveal references: Envelope labels are short,long,long,long. → Compare: Three of four agree; the disagreement is at 5 cm.

Point to each numbered stage and explain which recorded **input** it uses.

03. Frozen predictions versus reserved labels



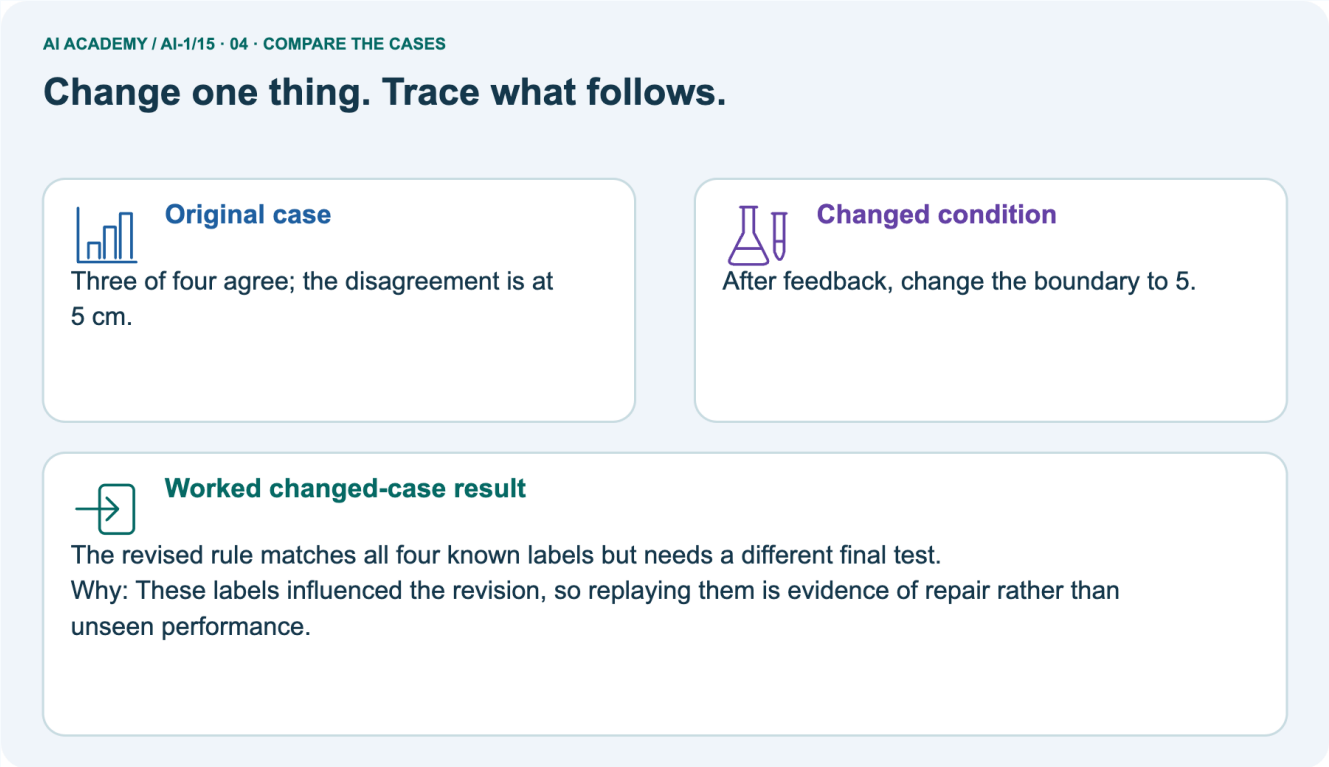
Follow the labelled picture with the matching explanation. Open the vector diagram

Frozen **prediction**s versus reserved labels. Test 1: Length cm = 3, PencilP = short, Reference = short, Result = agree. Test 2: Length cm = 5, PencilP = short, Reference = long, Result = disagree.

Test 3: Length cm = 6, PencilP = long, Reference = long, Result = agree. Test 4: Length cm = 9, PencilP = long, Reference = long, Result = agree.

Read the labels and values. Explain the relationship the model shows.

04. Change one thing. Trace what follows.



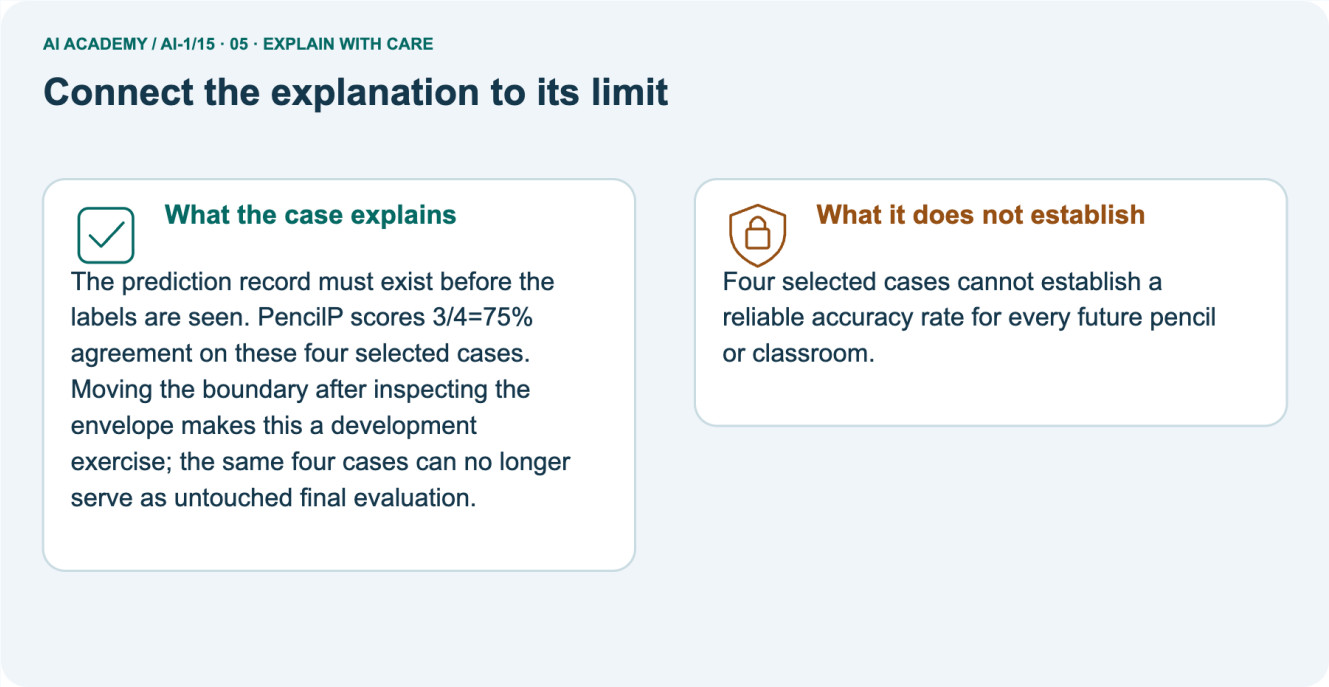
Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: Three of four agree; the disagreement is at 5 cm. Changed condition: After feedback, change the boundary to 5. Expected result: The revised rule matches all four known labels but needs a different final test.

Reason: These labels influenced the revision, so replaying them is **evidence** of repair rather than unseen performance.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

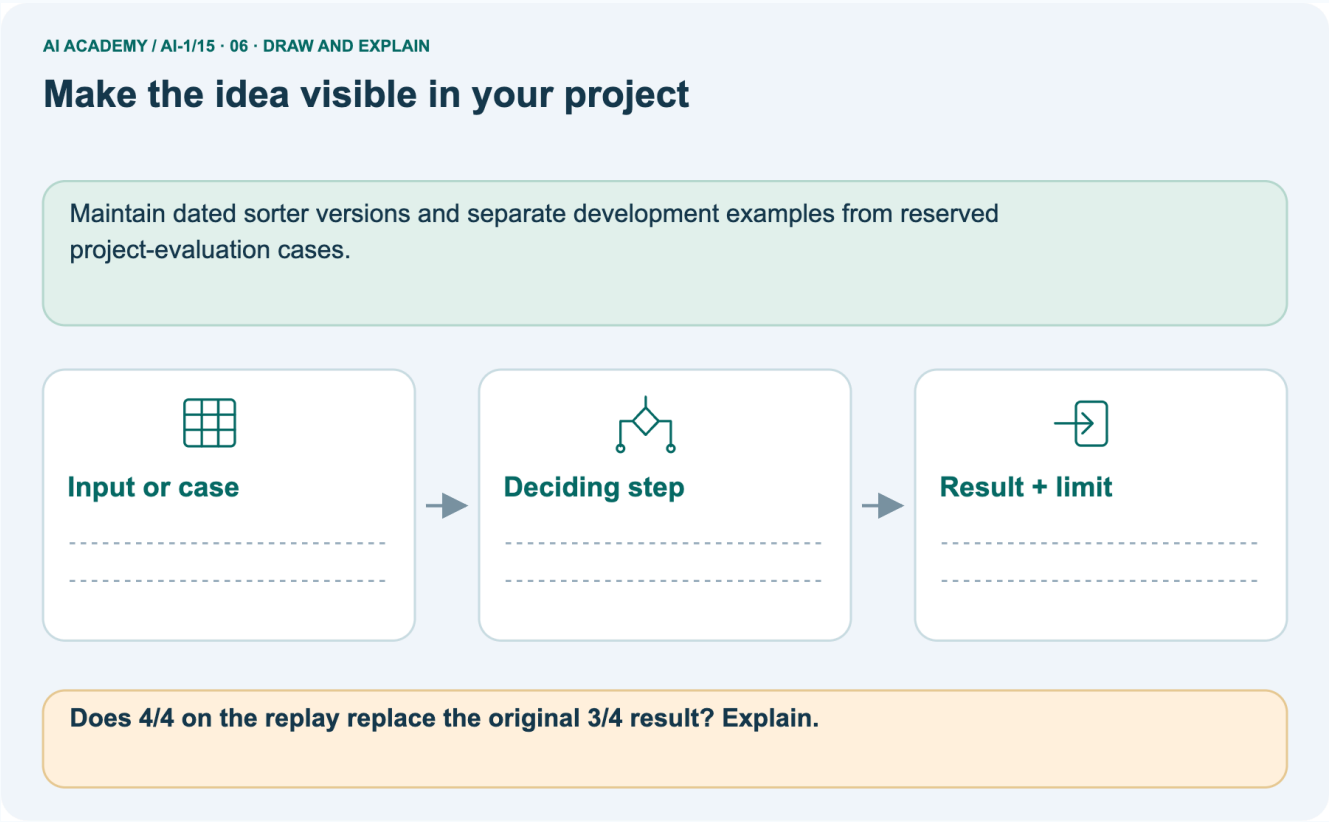
The **prediction** record must exist before the labels are seen. PencilP scores 3/4=75% agreement on these four selected cases.

Moving the boundary after inspecting the envelope makes this a development exercise; the same four cases can no longer serve as untouched final evaluation.

Limit: Four selected cases cannot establish a reliable accuracy rate for every future pencil or classroom.

Explain the case, then point to a limit on the claim.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Maintain dated sorter versions and separate **development examples** from reserved project-evaluation cases. Draw your **input**, deciding step, result and limit.

Supported question: Does 4/4 on the replay replace the original 3/4 result? Explain.

Draw your project trace and justify the deciding step using **evidence**.

▼ Earlier wording, alternatives and picture transcripts

01. Save New Examples for Testing: the case

02. Follow the deciding steps

03. Frozen predictions versus reserved labels

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.